CHRIST (Deemed to be University)
Department of Computer Science
Master of Computer Applications

Course: MCA412-2 - Research Methodology

LAB Exercise IV –

Dataset Identification and Description.

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Domain :

Marine Environmental Monitoring & Coastal Ecosystem-Preservation

Lab 4

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1. Dataset Identification

- Dataset: MARIDA: Marine Debris Archive from Sentinel-2 Satellite Data [cite: 10, 11]
- Repository & Link: Zenodo (https://doi.org/10.5281/zenodo.5151941) [cite: 10, 11]
- Publication: PLOS ONE (Kikaki et al., 2022; https://doi.org/10.1371/journal.pone.0262247) [cite: 10, 11]
- License: Creative Commons Attribution 4.0 International (CC BY 4.0) — the dataset is open-source for academic use as long as proper attribution is given [cite: 10, 11].

2. Source and Provenance

- Source: The dataset is curated on Zenodo through the EU OpenAIRE program and CERN infrastructure [cite: 10, 11].
- Acquisition: The data was extracted from European Space Agency (ESA) Sentinel-2 Multi-Spectral Instrument (MSI) Level-1C tiles. It covers 63 scenes from 11 countries collected between 2015 and 2021 [cite: 10, 11].
- Atmospheric Correction: The data was processed to Level-2A surface reflectance using the coastal ACOLITE processor [cite: 10, 11].

3. Data Type and Size

- Type: The dataset contains geospatial raster arrays (GeoTIFF, 32-bit float), vector masks, and CSV metadata [cite: 10, 11].
- Size & Coverage: It contains 1,387 image patches of 256 × 256 pixels at 10 m resolution. The dataset is approximately 3.4 GB when compressed and around 8.5 GB when uncompressed, with more than 830,000 expert-labeled pixels [cite: 10, 11].

4. Purpose and Key Variables

- Purpose: The dataset is mainly used as a benchmark for training and evaluating machine learning models for detecting marine litter while also distinguishing it from similar-looking materials [cite: 10, 11].
- Spectral Bands: It includes 11 Sentinel-2 MSI bands covering the Visible region (B1–B4), Red-Edge region (B5–B7, B8A), NIR (B8), and SWIR (B11, B12) [cite: 10, 11].
- Key Classes (15 categories): The main classes include Marine Debris (MD), Natural Organic Material (NOM), Sargassum (SA), Turbid Water (TW), Sediment-Dominated Water (SDW), Foam, Waves, and Ships [cite: 10, 11].
- Baseline Indices: The dataset also provides the Floating Debris Index (FDI), Floating Algae Index (FAI), NDVI, and NDWI [cite: 10, 11].

5. Relevance to Research Problem

- Target Isolation: The dataset provides separate ground-truth masks for Marine Debris (MD) and look-alike Natural Organic Material (NOM). This makes it useful for evaluating texture-based features and index-based discrimination [cite: 10, 11].
- Turbidity Baseline: It includes separate Turbid Water (TW) and Sediment-Dominated Water (SDW) classes, which can be used to test and calibrate turbidity-compensated detection algorithms [cite: 10, 11].
- Feature Extraction: The calibrated SWIR and NIR channels can be used for Gray-Level Co-occurrence Matrix (GLCM) spatial texture modelling and multi-band ratio tests [cite: 10, 11].

6. Limitations and Mitigation

- 10 m Pixel Resolution: The dataset captures aggregate patches rather than individual debris items. This limitation can be addressed using sub-pixel detection and spectral thresholding [cite: 10, 11].
- Class Imbalance: Marine debris represents around 1–2% of the total pixels. This can be handled using stratified sampling and class-weighted F1/IoU metrics [cite: 10, 11].
- High Turbidity Backscatter: Suspended sediments can increase the baseline NIR/SWIR reflectance. This can be addressed through the planned baseline offset compensation [cite: 10, 11].